ASSIGNMENT 1
Linear Algebra and Calculus
Unit 1: Introduction to Linear Algebra

Name: _____ **Roll No.:** _____

Class: _____ **Date:** _____

Instructions:

- Attempt all questions.
- Show all necessary steps of calculation.
- Give proper justification for conceptual questions.
- Use elementary row operations wherever required.

Part A:

Q1. Matrices

(a) Define a matrix. What is meant by the order of a matrix?

(b) Write down the order of the following matrices:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 2 & 3 \\ 4 & 5 \end{bmatrix}.$$

Q2. If

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ -2 & 3 \end{bmatrix},$$

find

$$A + B, \qquad A - B, \qquad 2A.$$

Q3. Equivalent Matrices

Define equivalent matrices. State the elementary row operations that can be performed on a matrix.

Q4. Elementary Matrix

What is an elementary matrix? Obtain the elementary matrix corresponding to the row operation

$$R_2 \to R_2 + 3R_1.$$

Q5. Row Reduction

Reduce the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 2 \end{bmatrix}$$

to row-echelon form.

Q6. Gauss–Jordan Reduction

Using Gauss–Jordan elimination, reduce

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

to the identity matrix.

Q7. Inverse of a Matrix

Find the inverse of

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

using the Gauss–Jordan method.

Q8. Linear Dependence

Determine whether the vectors

$$v_1 = (1, 2), \qquad v_2 = (2, 4)$$

are linearly dependent or linearly independent.

Q9. Rank

Find the rank of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 3 & 4 \end{bmatrix}.$$

Q10. System of Linear Equations

Determine whether the following system is consistent:

$$x + y = 5,$$

$$2x + 2y = 10.$$

Part B:

Q11. Reduce the matrix

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 3 & 2 \end{bmatrix}$$

to its **normal form** using elementary row and column operations.

Q12. Using Gauss–Jordan elimination, find A^{-1}, where

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Q13. Find the values of k for which the vectors

$$(1, 2, 3), \qquad (2, 4, k), \qquad (3, 6, 9)$$

are linearly dependent.

Q14. Determine whether the following vectors are linearly independent:

$$v_1 = (1, 0, 1), \qquad v_2 = (0, 1, 2), \qquad v_3 = (1, 1, 3).$$

Give a reason for your answer.

Q15. Find the rank of

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 1 & 2 & 3 \\ 3 & 5 & 8 & 11 \end{bmatrix}.$$

Q16. Solve the following system using Gauss–Jordan elimination:

$$x + y + z = 6,$$

$$2x - y + z = 3,$$

$$x + 2y - z = 2.$$

Q17. Determine whether the system

$$x + y + z = 3,$$

$$2x + 2y + 2z = 6,$$

$$x + y + z = 4$$

has a solution.

Justify your answer using the ranks of the coefficient matrix and the augmented matrix.

Q18. Find the values of k for which the system

$$x + y + z = 2,$$

$$2x + 3y + z = 5,$$

$$x + 2y + kz = 4$$

has a unique solution.

Part C:

Q19. A student says:

> "If two matrices have the same rank, then they must be equivalent."

Is the statement true or false? Give a suitable explanation.

Q20. Consider

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

A student says that A is invertible because all its entries are non-zero.

Do you agree? Explain your answer without directly using the determinant formula.

Q21. Without completely solving the system, determine whether

$$x + y + z = 3,$$

$$2x + 2y + 2z = 6,$$

$$3x + 3y + 3z = 9$$

has a unique solution, infinitely many solutions, or no solution.

Give a reason based on rank.

Q22. Can a matrix have rank greater than its number of rows? Explain with a reason.

Q23. Suppose A is a 3×3 matrix and

$$\text{rank}(A) = 2.$$

Answer the following:

(a) Can A^{-1} exist?

(b) Are the columns of A linearly independent?

(c) What can you say about the system $AX = 0$?

(d) What is the rank of the row-reduced form of A?

Q24. Consider the vectors

$$v_1 = (1, 2, 3), \qquad v_2 = (2, 4, 6), \qquad v_3 = (1, 1, 1).$$

A student says:

> "Since there are three vectors in $\mathbb{R}^3$, they must be linearly independent."

Check whether the student's reasoning is correct. Give a proper justification.

Q25. Find the value of k for which the following system has **infinitely many solutions**:

$$x + y + z = 3,$$

$$2x + 2y + 2z = 6,$$

$$x + 2y + kz = 5.$$

Q26. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$$

Without finding A^{-1}, explain why the inverse does not exist.

Then find the **rank** of A and determine whether its columns are linearly dependent or linearly independent.

— End of Assignment —